ONLINE UPS SIMULATION

OrCAD and MATLAB/Simulink

Name: Aarnav Rao,Mayank Pal Roll No.: 24BEE0079,24BEE0103

Subject:Power Electronics Date: 28/10/2026

1. Aim

To study and simulate an Uninterruptible Power Supply (UPS) using two approaches: (i) a circuit-level model in OrCAD/PSpice and (ii) a signal-level functional model in MATLAB/Simulink. The study focuses on AC input, rectification/DC-link representation, inverter operation, output filtering, and the expected continuity of AC output when the mains supply is unavailable.

2. Apparatus / Software Required

- Computer or laptop capable of running the simulation software.
- OrCAD Capture / PSpice for schematic entry and circuit simulation.
- MATLAB/Simulink for the signal-level UPS model.
- OrCAD circuit components shown in the schematic: 230 V AC, 50 Hz source; 1N4007 diodes; MOSFET models; inductors; capacitors; resistors; pulse/gate-drive sources; and controlled switches.
- Simulink blocks: Sine Wave, Abs, Gain, Transfer Fcn, Sum, Saturation, Switch, Step/Constant, subsystem Inports/Outports, and Scope blocks.

3. Theory

3.1 Introduction

An Uninterruptible Power Supply (UPS) provides power to a load when the normal mains supply is interrupted or becomes unsuitable. A typical online UPS continuously converts the incoming AC to DC and then uses an inverter to generate the AC output. A battery or battery-backed DC source supports the DC link during a mains failure. This arrangement can provide a continuous output because the load is normally supplied by the inverter.

The two models in this report serve different purposes. The OrCAD schematic represents an electrical circuit with diode, MOSFET, passive-filter and load elements. The MATLAB/Simulink model is a simplified signal-level representation built from standard Simulink blocks; it is useful for illustrating the functional signal path and control logic, but it does not reproduce all semiconductor switching, battery electrochemistry, or physical power-stage behaviour.

3.2 Circuit Description — OrCAD

The OrCAD schematic shows a 230 V AC, 50 Hz input source. An input EMI inductor and capacitors are placed ahead of a four-diode bridge rectifier using 1N4007 diode models. The rectifier produces a unidirectional pulsating DC waveform. The downstream stage uses MOSFET models with pulse sources as gate-drive signals, together with an inductor and capacitor to form a switching/conversion and filtering stage. Controlled switches and a resistive load are included in the output portion of the schematic.

The displayed OrCAD waveform is the voltage measured across the load resistor. It is approximately periodic at 50 Hz and resembles an AC sinusoid, although small distortions are visible around some transitions. The schematic and

waveform are reproduced as provided; detailed performance claims such as regulation, efficiency, battery runtime, and successful mains-failure transfer cannot be established from these two screenshots alone.

3.3 Circuit Description — MATLAB/Simulink

The Simulink model is organized into subsystems, including AC_Input_Monitor, Rectifier_Model, DC_Bus_Controller, BMS, UPS_Controller, DC_Source_Select, and SPWM_Inverter. The rectifier subsystem processes the AC signal using an absolute-value operation and a transfer function/gain chain. The DC-source selection block chooses between the rectifier signal and a 400 V constant used as an idealized battery-backed DC source. The inverter subsystem generates an AC reference signal and also produces a separate PWM pulse signal for monitoring. An output filter smooths the modeled AC waveform.

Because the model uses standard Simulink signal blocks rather than Simscape electrical components, its 400 V battery-backed source is an ideal signal placeholder. The model is therefore a functional demonstration of UPS signal flow, not a detailed physical battery or power-electronics simulation.

3.4 Relevant Equations

For the 230 V RMS, 50 Hz AC input shown in the OrCAD schematic:

$V_{in}(t) = V_{peak} \sin(2\pi f t)$, where $V_{peak} = \sqrt{2}\, V_{RMS} \approx \sqrt{2} \times 230 \approx 325$ V

Thus, $V_{in}(t) \approx 325 \sin(2\pi \times 50\, t)$ V.

The input period is $T = 1/f = 1/50 = 0.02$ s = 20 ms.

For an ideal full-wave bridge rectifier, the unfiltered output is approximately $V_{rect}(t) = |V_{in}(t)|$, less the forward-voltage drops of the two conducting diodes. The ripple repetition frequency is $2f = 100$ Hz for a 50 Hz input.

For the signal-level Simulink inverter, the modeled output is formed approximately as $V_{out}(t) = V_{DC} \times m \sin(2\pi f t)$, where V_{DC} is the selected DC-link signal and m is the chosen reference amplitude factor. This expression describes the simplified signal-generation approach; it is not a full switched-inverter equation.

3.5 Waveform Interpretation

The OrCAD output plot spans 0 to 100 ms and shows approximately five cycles, consistent with a frequency near 50 Hz. Its peaks are approximately +320 V and −320 V. The waveform is close to sinusoidal but contains visible distortion near parts of the cycle.

The MATLAB/Simulink final-output plot also spans 0 to 0.1 s and shows approximately five cycles. Its peak values are approximately +300 V and −300 V, giving an approximate peak-to-peak range of 600 V. The plotted period is about 20 ms, corresponding to 50 Hz. These are readings estimated from the screenshots, not precision measurements.

4. Software Screenshots and Output Waveforms

4.1 OrCAD Circuit Schematic

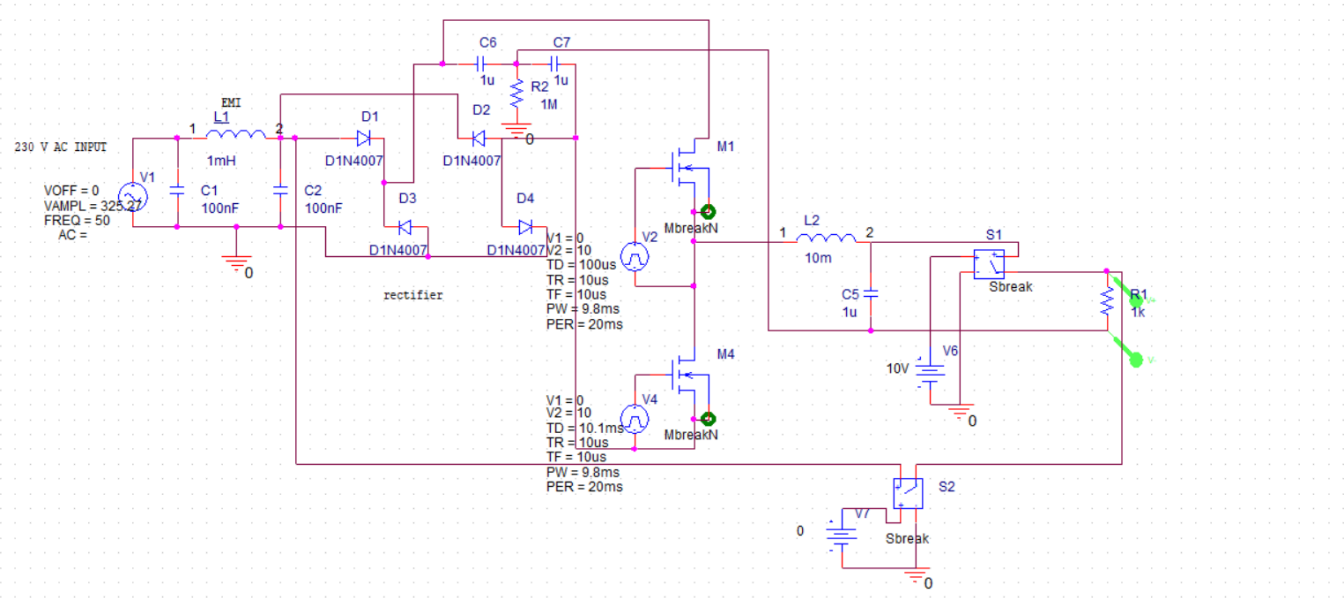


Figure 1. OrCAD/PSpice UPS circuit schematic supplied for the report.

4.2 OrCAD Output Waveform

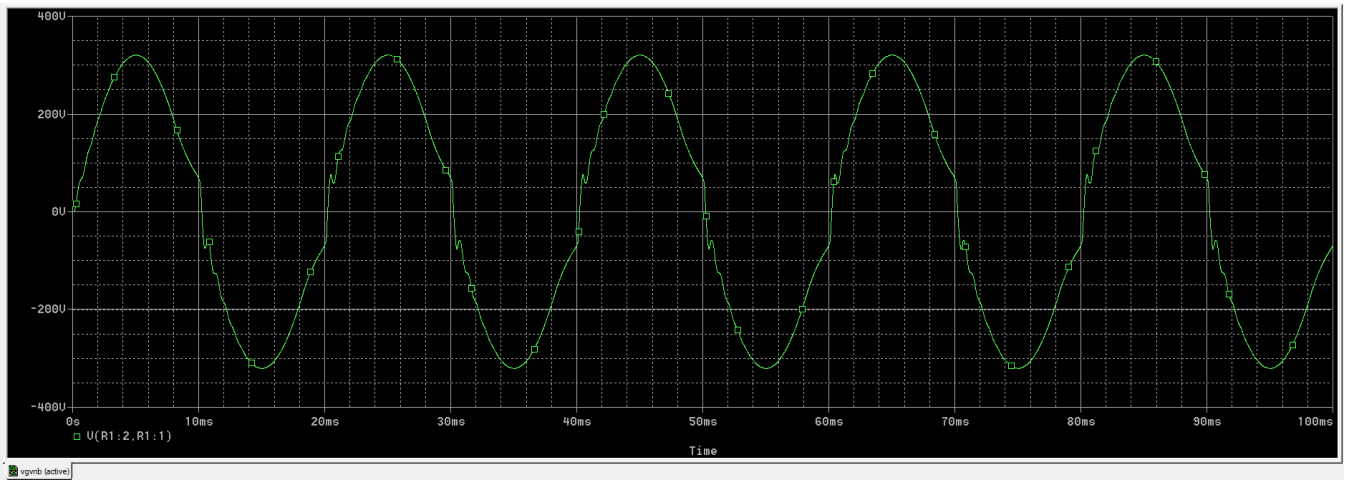


Figure 2. Simulated load-voltage waveform from OrCAD/PSpice.

4.3 MATLAB/Simulink Model — Rectifier and DC-Bus Section

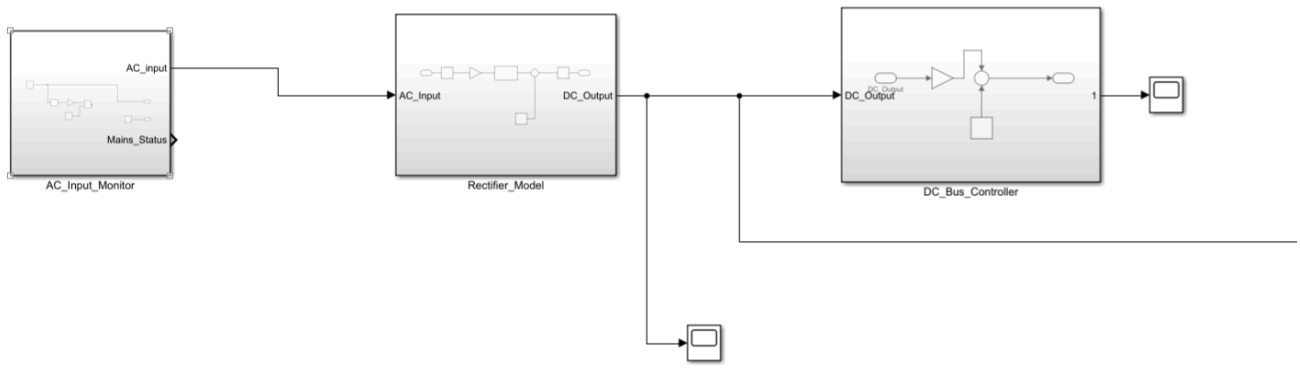


Figure 3. MATLAB/Simulink model showing the AC input monitor, rectifier model and DC-bus controller.

4.4 MATLAB/Simulink Model — BMS, Controller and Inverter

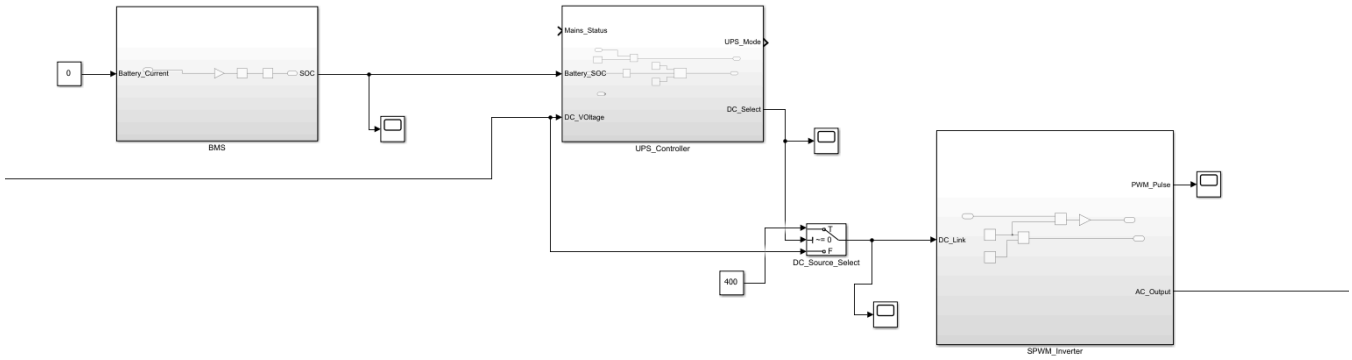


Figure 4. MATLAB/Simulink model showing the BMS, UPS controller, DC-source selector and SPWM inverter.

4.5 MATLAB/Simulink AC Output Waveform

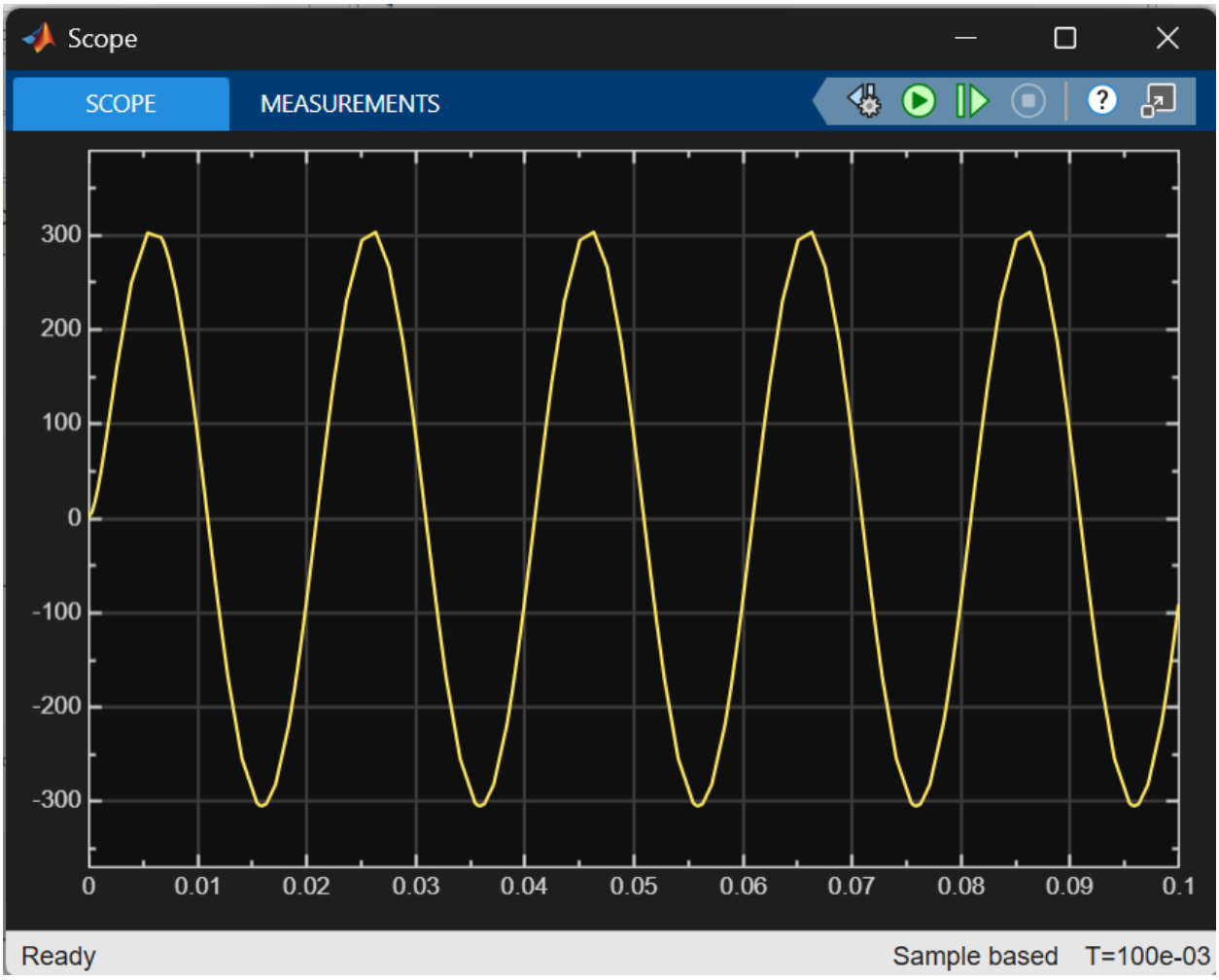


Figure 5. Final filtered AC output waveform from MATLAB/Simulink (approximately 50 Hz, ±300 V peak).

5. Conclusion

The project demonstrates two simulation approaches for studying UPS operation. The OrCAD model presents a circuit-level arrangement containing a bridge rectifier, MOSFET switching stage, passive components and a load. Its simulated load waveform is approximately sinusoidal at 50 Hz. The MATLAB/Simulink model provides a modular signal-level representation of rectification, DC-source selection, battery-status/control logic, inverter signal generation and output filtering. The Simulink output waveform is also approximately sinusoidal at 50 Hz.

The models are not directly equivalent in physical detail: the Simulink model uses an ideal 400 V constant for the battery-backed source and does not simulate battery chemistry or the full switched power stage. Therefore, successful real-world regulation, backup runtime, efficiency and seamless transfer would require further validation and, for a physical model, more detailed component and control modelling.

6. Result

The UPS models were assembled and simulated in OrCAD/PSpice and MATLAB/Simulink. The provided output plots show approximately 50 Hz AC waveforms with peak voltages of about ±320 V in OrCAD and ±300 V in MATLAB/Simulink. The Simulink model demonstrates the intended functional signal path, including a nominal 400 V idealized DC-link/battery-backed source. The stated results are based on the supplied simulation screenshots.

7. Comparative Analysis of UPS Simulation Using OrCAD and MATLAB/Simulink

7.1 Introduction

The UPS was studied using two simulation platforms: OrCAD/PSpice and MATLAB/Simulink. OrCAD was used for a circuit-level representation containing electronic components, while the MATLAB/Simulink model represents the main UPS functions through interconnected signal-processing and control blocks. The platforms therefore provide complementary views of UPS operation rather than identical implementations.

7.2 Comparative Analysis

Parameter	OrCAD/PSpice	MATLAB/Simulink
Modelling approach	Circuit-level simulation using electrical component models.	Signal-level functional modelling using interconnected Simulink blocks.
AC input	230 V AC, 50 Hz source shown in the schematic.	Sine Wave block represents the AC input signal.
Rectification	Four-diode bridge rectifier using 1N4007 diode models.	Abs, Transfer Fcn and Gain blocks approximate rectification and smoothing.
Switching/inverter stage	MOSFET models with pulse sources and passive components.	Signal-based inverter waveform generation with a separately monitored PWM signal.
DC backup source	Depends on the source and switching arrangement in the circuit.	A Constant block of 400 V is used as an idealized backup DC source.
Output filtering	Inductors and capacitors in the electrical circuit.	Signal-processing/filter blocks smooth the modelled output.
Observation method	Voltage can be measured at circuit nodes; the supplied plot shows load voltage.	Scope blocks display internal signals, PWM pulses and final AC output.
Main strength	Useful for examining circuit topology and component-level electrical behaviour.	Useful for explaining functional stages, control logic and signal flow.
Main limitation	Results depend on component models, circuit parameters and simulation settings.	The current signal-level model does not fully represent physical switching, battery dynamics or power losses.

7.3 Comparison of Output Waveforms

The supplied OrCAD output waveform covers approximately 0 to 100 ms and displays about five cycles, indicating a frequency near 50 Hz. Its peak values are approximately +320 V and −320 V, with some visible distortion around portions of the waveform.

The supplied MATLAB/Simulink output waveform also displays about five cycles over 0.1 s, corresponding to approximately 50 Hz. The observed peaks are approximately +300 V and −300 V. These peak values are estimates from the screenshots, and the difference should not be treated as a direct accuracy comparison because the models use different implementations and may not share identical measurement conditions.

7.4 Overall Comparative Assessment

OrCAD/PSpice is more suitable for investigating circuit-level behaviour when appropriate electrical component models are used. The developed MATLAB/Simulink model is more suitable for demonstrating the functional UPS architecture, source-selection logic, and signal flow. The Simulink implementation uses an idealized 400 V backup signal and simplified inverter processing; therefore, it does not by itself validate physical battery runtime, conversion efficiency, semiconductor stress, or real-world transfer performance. The two models are complementary and should not be described as electrically equivalent.

8. Comparative Conclusion

The comparative study shows that OrCAD/PSpice and MATLAB/Simulink provide complementary methods for studying UPS operation. OrCAD represents a component-based electrical circuit and is useful for examining circuit nodes and the effects of the rectifier, switching devices, passive components and load. MATLAB/Simulink provides a modular signal-level representation that makes the functional stages, control signals and output waveform easier to trace. Both supplied output plots show an approximately 50 Hz AC waveform, although the estimated peak values differ. Because the Simulink model uses an idealized backup source and simplified signal processing, its results demonstrate functional behaviour rather than full physical validation. A detailed switching-based Simulink model and matched test conditions would be needed for a rigorous quantitative comparison.

9. Comparative Result

The comparative analysis was completed using the supplied OrCAD/PSpice and MATLAB/Simulink models and their output waveforms. The OrCAD plot shows an approximately 50 Hz waveform with peaks near ±320 V, while the Simulink plot shows an approximately 50 Hz waveform with peaks near ±300 V. OrCAD demonstrates circuit-level simulation, whereas the current Simulink model demonstrates the intended signal-level UPS functions, including DC-source selection and inverter output generation. The results confirm the usefulness of both tools for different aspects of UPS study; they do not establish equivalent physical performance.